npj | materials degradation

Article

Published in partnership with CSCP and USTB

https://doi.org/10.1038/s41529-026-00746-3

Enhancement mechanisms of Cr and RE on the corrosion resistance of HRB400 rebar in chloride-containing concrete pore solution

Check for updates

Renzheng Zhu[1,2,4], Tianqi Chen[1,2,4], Lianjun Hao[1,2], Weiyong Yang[3], Changyou Gao[3], Xiaotan Zuo[3], Xuequn Cheng[1,2], Chao Liu[1,2] ✉ & Xiaogang Li[1,2] ✉

This study deciphers the chloride-induced early-stage corrosion of Cr/RE-microalloyed HRB400 rebars in saturated $Ca(OH)_2$ pore solutions (0.5–3.5 wt% Cl^-). Rare-earth addition transforms original MnS and Al_2O_3–MnO–CaO inclusions into RE–Al–O–S particles, many encapsulated by thin MnS shells. Preferential MnS dissolution creates occluded cells whose acidification simultaneously attacks the exposed RE–Al–O core and adjacent steel, yet the RE phase markedly retards this sequence. Across all chloride levels, HRB400-Cr-RE exhibits the highest pitting potential, lowest passive current density, largest charge-transfer resistance and thinnest passive-film donor density; after 7 d immersion its corrosion rate is one-third that of HRB400 and confocal microscopy confirms the shallowest pits with the lowest aspect ratios. Elevated Cl^- progressively lowers pitting potentials, raises passive currents, shrinks Nyquist arcs and increases donor densities in all steels, evidencing accelerated dissolution of inclusions and surrounding matrix under high-chloride conditions.

Reinforced concrete is vital for infrastructure projects like bridges, docks, tunnels, and subways, owing to its excellent cost-performance ratio and mechanical strength. It offers exceptional resistance to various environmental stresses, including abrasion, chemical attacks, and fatigue. However, its long-term durability, critical for serviceability, is compromised by combined deterioration mechanisms such as chloride-induced corrosion, carbonation, alkali-silica reactions, and freeze-thaw cycles[1–3]. Among these, steel reinforcement corrosion in highly aggressive environments—particularly marine settings with high salinity, humidity, and wet-dry cycles—remains a persistent challenge despite extensive research and mitigation efforts[4–6].

In marine environments, chloride ingress represents the predominant degradation mechanism, accounting for over 60% of reinforced concrete failures[7]. Chloride ions—either intrinsic to construction materials or externally derived from seawater—diffuse through the concrete matrix via capillary action and permeation, ultimately accumulating at the steel-concrete interface[8,9]. While numerous conventional approaches have been investigated to mitigate chloride attack—including enhancing concrete density[10], increasing cover thickness, applying protective coatings[11], employing electrochemical techniques[12], and modifying reinforcement surface characteristics[13,14]—these strategies fundamentally modify the steel's external environment rather than its intrinsic corrosion resistance. This critical limitation inherently constrains their long-term protective effectiveness and durability, particularly in harsh marine conditions. A more fundamental solution lies in alloy design and microstructural optimization of the steel itself[15–17]. By enhancing the material's inherent corrosion resistance through compositional and microstructural engineering, this approach offers a more sustainable and potentially more effective strategy for combating chloride-induced corrosion in marine concrete structures, addressing the root cause rather than merely mitigating symptoms.

Recent studies[18–22] have provided detailed insight into how MnS inclusions initiate localized corrosion in carbon and stainless steels. MnS tends to dissolve preferentially, creating micro-galvanic cells between the inclusion and surrounding matrix and triggering early pitting. These works collectively highlight that inclusion chemistry and the local electrochemical environment govern pit initiation and propagation. Rare earth (RE)

[1]Key Laboratory for Corrosion and Protection of the Ministry of Education, Institute of Advanced Materials & Technology, University of Science and Technology Beijing, Beijing, China. [2]National Materials Corrosion and Protection Data Center, University of Science and Technology Beijing, Beijing, China. [3]Wuhu Xinxing Ductile Iron Pipe Co., Ltd, Hefei, Anhui, China. [4]These authors contributed equally: Renzheng Zhu, Tianqi Chen. ✉e-mail: liuchao@ustb.edu.cn; lixiaogang@ustb.edu.cn

elements serve as potent alloying additives that enhance steel purity by refining non-metallic inclusions, thereby improving the pitting corrosion resistance of rebars. Beyond inclusion control, RE elements promote uniform matrix dissolution and participate in corrosion product formation, facilitating a denser, more protective rust layer. Chromium (Cr), on the other hand, is a well-known corrosion-resistant element that plays a critical role in forming a stable passive oxide layer on steel surfaces. This passive layer acts as a protective barrier, preventing the penetration of corrosive agents, particularly in chloride-contaminated environments. Cr also enhances the steel's resistance to localized corrosion such as pitting, which is a common issue in concrete reinforcement exposed to aggressive chloride ions. Although both RE and Cr individually enhance the corrosion resistance of rebars, their synergistic effects have not been fully explored. The interaction between RE and Cr may result in more complex and effective corrosion protection mechanisms, offering superior performance compared to using either element alone. Addressing this knowledge gap is crucial for developing advanced corrosion-resistant rebar materials, especially in harsh environments with high chloride concentrations.

This study systematically investigates Cr/RE-regulated inclusion effects on microstructural evolution, electrochemical behavior, and passive film properties through comprehensive electrochemical testing and immersion experiments in chloride-containing SCP solutions. By elucidating the fundamental mechanisms of Cr/RE-enhanced corrosion resistance, this work provides critical insights for designing durable reinforcement materials for marine applications.

Results

Microstructure characterization

Figure 1 presents the metallographic structures of the three steel rebar specimens, revealing consistent microstructural features dominated by ferrite-pearlite constituents with minor granular bainite, all exhibiting characteristic banded morphology. Chromium's strong ferrite-stabilizing effect significantly reduced pearlite content, while rare earth additions modified carbon activity in ferrite, increasing its carbon solubility and consequently altering the ferrite/pearlite ratio.

Figure 2 shows the morphological and compositional evolution of inclusions in HRB400 steel following Cr and RE additions. Untreated HRB400 and Cr-modified samples contained irregular MnS and Al_2O_3 inclusions along with (Ca,Mg,Al)-O composites (Fig. 2a, b). These inclusions exhibited angular geometries (spherical, reticular, or elongated) with sharp edges and sizes <2 μm—features known to initiate microcracks and degrade both mechanical properties and pitting resistance[23–25]. The types of inclusions in HRB400-Cr-RE steel bar with RE addition are $REAlO_3$ and $REAlO_3$ inclusions wrapped by MnS, and the single MnS inclusions disappear.

To quantify the effect of alloying on inclusion characteristics, a statistical analysis of inclusion size distribution was conducted based on more than 100 inclusions per sample. As shown in Fig. S1, Cr and RE additions markedly refined the inclusions: the fraction of inclusions smaller than 2 μm increased sharply, while that of inclusions larger than 5 μm decreased. This refinement contributes to a more uniform inclusion dispersion and improved interfacial stability, which are beneficial for corrosion resistance.

Potentiodynamic polarization curves

Figure 3 presents the potentiodynamic polarization (PDP) curves of the three reinforcing steels in saturated $Ca(OH)_2$ simulated concrete pore solution (SCP) with increasing NaCl concentrations, while Table 1 compiles the corresponding pitting potentials (E_{pit}) and passive current densities (i_{pass}). Irrespective of alloy type, the anodic branches in Fig. 3a–c enter the passive region directly, without a discernible active–passive transition, and the virtually superimposed cathodic branches indicate that oxygen reduction and hydrogen evolution jointly govern the cathodic kinetics. The congruent profiles imply that the underlying corrosion mechanism remains unchanged despite the Cr/RE alloying; only the kinetics are altered.

In SCP containing 0.5%, 1%, and 2% Cl^-, all electrodes anodically passivate, whereas the cathodic reaction remains charge-transfer controlled. The quantitative data in Table 1 reveal a pronounced alloying benefit. At low chloride concentrations (0.5 wt.%), the passive films formed on both steels remain stable, and the corrosion process is mainly controlled by the passive film's breakdown potential rather than by inclusion activity. Under these mild conditions, the Cr–RE effect is less pronounced. At 1 wt.% Cl^- (Fig. 3b), E_{pit} rises from 292 mV for HRB400 to 446 mV for HRB400-Cr and 654 mV for HRB400-Cr-RE, while i_{pass} falls concomitantly. Similar trends persist at 2 wt.% Cl^- (Fig. 3c); HRB400-Cr-RE exhibits the lowest i_{pass} (11.3 μA cm^{-2}), one-third that of HRB400 (38 μA cm^{-2}). At 3.5 wt% Cl^- (Fig. 3d), the curves lose the sharp passivation jump, displaying instead a sluggish active-to-passive transition that signals severe attack on the passive film. Even under this aggressive condition, HRB400-Cr-RE retains the highest E_{pit} and the lowest i_{pass} (Table 1), underscoring the potent synergy of Cr and RE in suppressing chloride-induced corrosion.

Electrochemical impedance spectroscopy

Figure 4 presents the impedance spectra (including Nyquist and Bode) for the three rebar samples exposed to chloride-containing SCP. All Nyquist traces show single, slightly depressed capacitive arcs, confirming that Cr/RE micro-alloying modifies only the kinetics while preserving the interfacial mechanism. The arc radii follow the order HRB400 < HRB400-Cr < HRB400-Cr-RE, indicating a progressive improvement in charge-transfer resistance due to the influence of Cr and RE.

Although the Bode plots show a single dominant time constant, a weak high-frequency shoulder in the Nyquist plots indicates the contribution of a surface film. Hence, a two-time-constant equivalent circuit was employed to describe both the film and charge-transfer processes. HRB400-Cr-RE attains phase angles close to 80°, whereas HRB400 barely reaches 70°, and the impedance modulus follows the same trend. Figure 4b–d reveal that elevating Cl^- from 1 wt% to 3.5 wt% shrinks the arcs, yet the ranking HRB400-Cr-RE > HRB400-Cr > HRB400 remains intact. Consequently, Cr/RE alloying systematically enlarges interfacial resistance and markedly improves chloride tolerance.

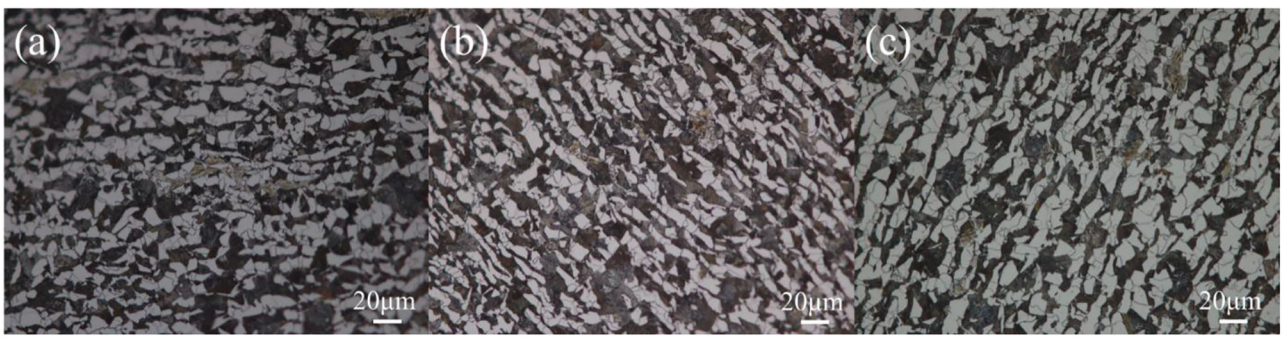


Fig. 1 | Micrometallurgical structure of three samples. a HRB400; **b** HRB400-Cr; **c** HRB400-Cr-RE.

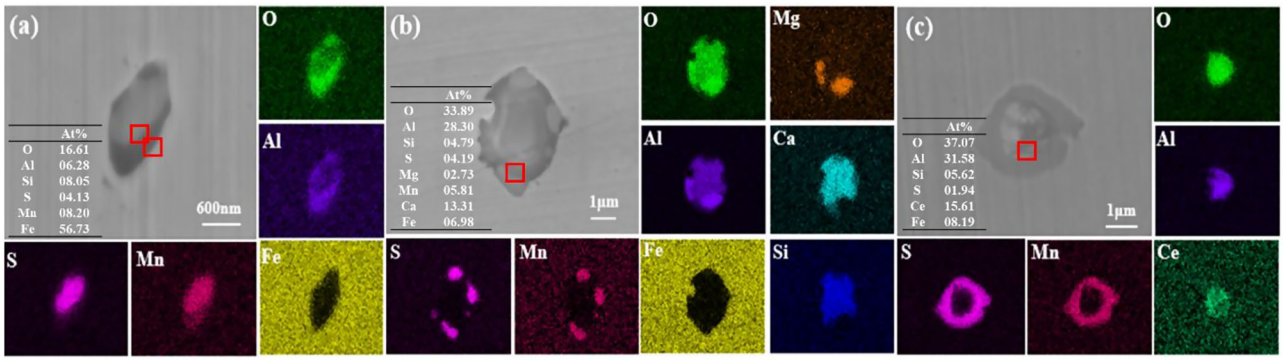


Fig. 2 | SEM/EDS results of inclusions across the three steels. a Al_2O_3–MnS clusters together with Al_2O_3–MnO–CaO phases in HRB400; **b** HRB400-Cr is (Ca, Mg, Al)–O complex oxides in HRB400-Cr; **c** RE–Al–O enveloped by thin MnS shells in HRB400-Cr-RE.

Fig. 3 | The PDP curves of three rebars in SCP solution containing different concentrations of sodium chloride. a 0.5 wt% NaCl; **b** 1 wt% NaCl; **c** 2 wt% NaCl; **d** 3.5 wt% NaCl.

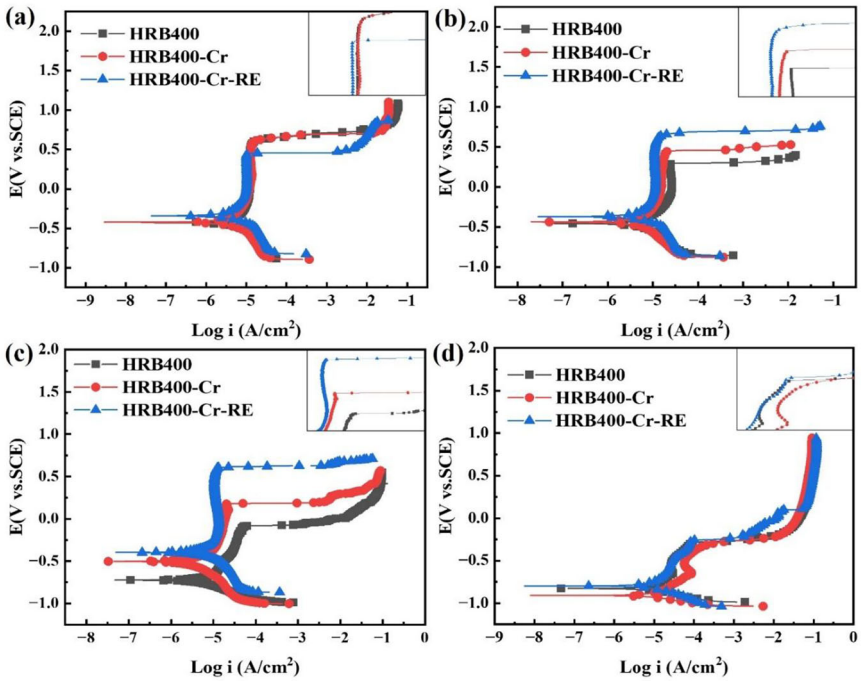


Table 1 | Electrochemical parameters of three kinds of steel electrodes in SCP solution containing different concentrations of sodium chloride

Cl^- (wt%)	E_{pass} (mV_{SCE})			I_{pass} (μA/cm²)		
	HRB400	HRB400-Cr	HRB400-Cr-RE	HRB400	HRB400-Cr	HRB400-Cr-RE
0.5	625	618	431	12.6	12.0	10.0
1	292	446	654	25.7	17.4	10.7
2	78	189	615	38.0	20.9	11.3
3.5	−292	−299	−255	72.4	107.2	37.2

Impedance spectra were analysed with ZSimpWin by fitting the equivalent circuit shown in Fig. 4i; the model describes all interfacial physicochemical processes and yields excellent agreement with the experimental data ($\chi^2 < 0.01$), in line with earlier work[26]. The extracted parameters in Table 2 are defined as follows: R_s reflects the ohmic resistance of the bulk solution; R_f and Q_f describe the barrier properties and non-ideal capacitance of the surface film; R_{ct} and Q_{dl} quantify the electron-transfer resistance and the double-layer response at the metal–electrolyte interface. A constant-phase element was substituted for an ideal capacitor to accommodate the distributed capacitance arising from surface heterogeneity, as outlined in Eq. (1):

$$Z_{CPE} = \frac{1}{Y_0(j\omega)^n} \quad (1)$$

In the expression, Y_0 denotes the CPE constant, ω the angular frequency, $j^2 = -1$, and n the dispersion parameter. A value of $n = 1$ signifies

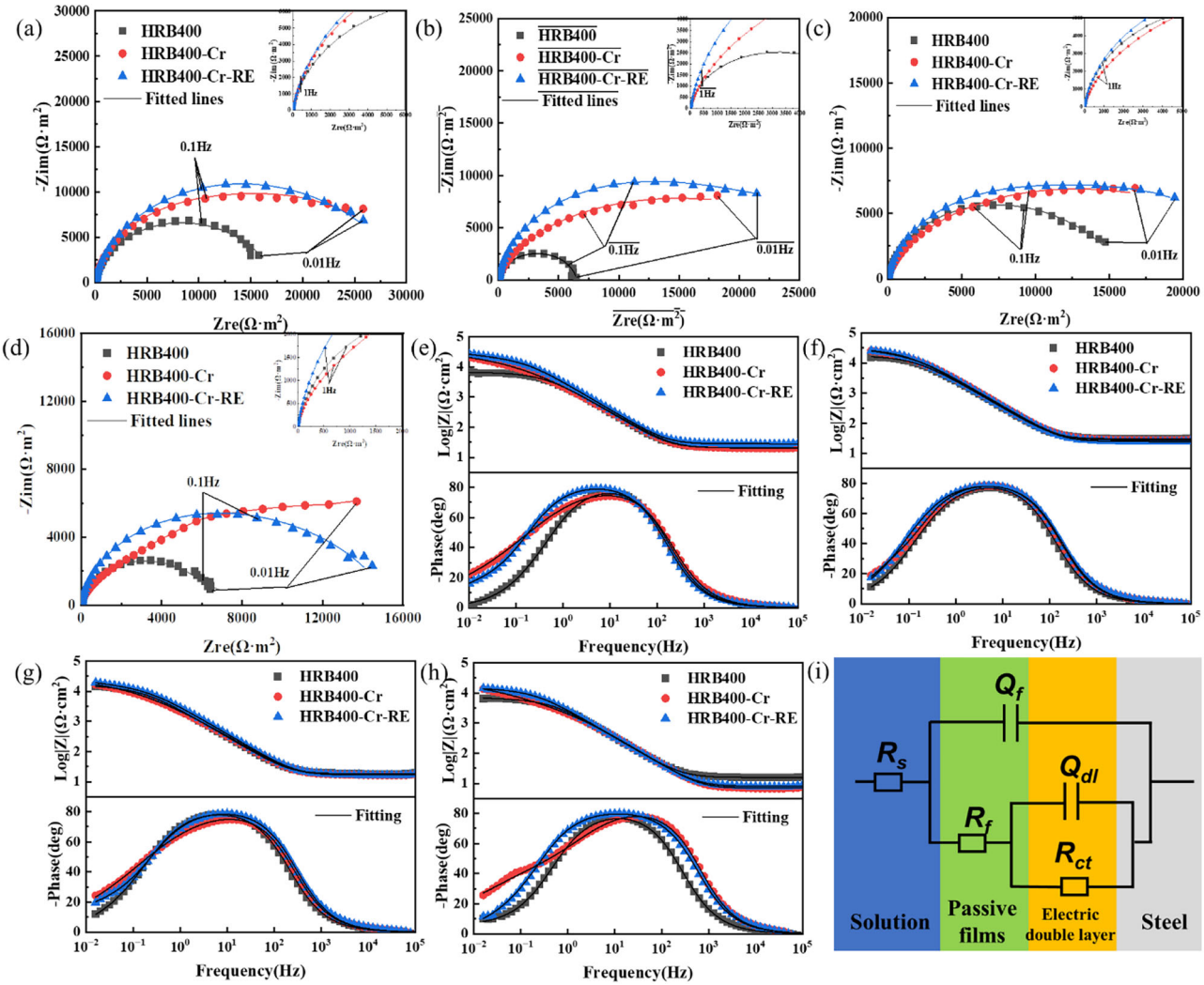


Fig. 4 | EIS spectra of in SCP solution containing different concentrations of sodium chloride. a~**d** Nyquist diagram, **e**~**h** Bode diagram, **a**, **e** 0.5 wt% NaCl; **b**, **f** 1 wt% NaCl; **c**, **g** 2 wt% NaCl; **d**, **h** 3.5 wt% NaCl. **i** Equivalent circuits used to model the EIS spectra.

Table 2 | Fitting parameters of the EIS results of three kinds of steel electrodes in SCP solution containing different concentrations of sodium chloride

Cl^-(wt.%)	Sample	R_s,	R_f,	Q_{dl}		R_{ct},	Q_f		R_p
		Ω cm²	kΩ cm²	Y_0 (10^{-4} Ω^{-1} cm^{-2} s^n)	n	kΩ cm²	Y_0 (10^{-4} Ω^{-1} cm^{-2} sn)	n	kΩ cm²
0.5	HRB400	29.9	10.1	0.70	0.92	7.8	1.13	0.54	17.9
	HRB400-Cr	27.5	18.4	0.62	0.91	10.8	3.34	0.83	29.2
	HRB400-Cr-RE	24.5	18.8	0.67	0.91	11.6	1.46	0.58	30.4
1	HRB400	20.7	5.2	0.71	0.92	5.2	0.61	0.88	10.4
	HRB400-Cr	20.3	3.7	0.72	0.90	32.6	0.94	0.43	36.3
	HRB400-Cr-RE	27.9	16.4	0.57	0.93	21.4	0.22	0.53	37.8
2	HRB400	18.2	10.2	0.74	0.92	5.9	2.74	0.63	16.1
	HRB400-Cr	16.6	2.5	0.69	0.91	26.8	0.90	0.45	29.3
	HRB400-Cr-RE	16.7	10.8	0.59	0.92	24.7	1.90	0.48	35.5
3.5	HRB400	15.7	4.7	0.65	0.92	2.5	3.10	0.57	7.2
	HRB400-Cr	6.9	1.8	0.57	0.94	4.5	1.55	0.51	15.2
	HRB400-Cr-RE	7.9	10.7	0.65	0.92	26.7	4.09	0.68	28.5

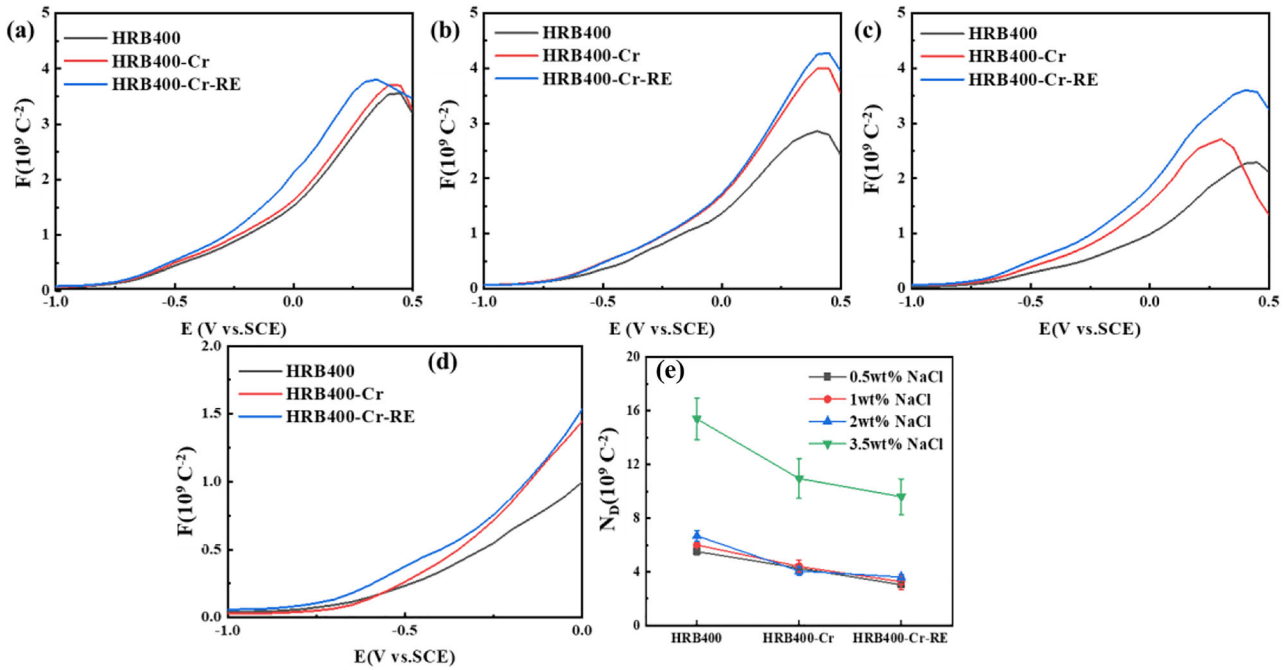


Fig. 5 | M-S curves in SCP solution containing different concentrations of NaCl. a 0.5 wt% NaCl; **b** 1 wt% NaCl; **c** 2 wt% NaCl; **d** 3.5 wt% NaCl **e** calculated carrier density (N_D).

ideal capacitive behaviour, $n = 0$ indicates pure resistance, and $n = -1$ corresponds to inductive response, whereas $0 < n < 1$ points to interfacial heterogeneity.

The R_s values were verified after extending the frequency range to 100 kHz and were found to be stable. The constant-phase element Q_f was converted to an effective film capacitance C_f using the Brug equation. The dispersion coefficient indicates a mixed capacitive–diffusive response associated with ionic transport through the Ce/La-enriched passive film. Table 2 reveals that introducing Cr and RE raises R_{ct} from 7.8 kΩ cm^2 in plain HRB400 to 11.6 kΩ cm^2 in HRB400-Cr-RE under 0.5 wt% Cl^-–SCP, indicating a more compact and Cr/RE-enriched surface layer that outperforms conventional iron oxides in retarding charge transfer[27]. Furthermore, the overall corrosion resistance can be gauged by the polarization resistance, Rp, defined here as the sum of R_{ct} and Rf[28]. Mirroring the Nyquist trends, the polarization resistance (R_p) consistently ranks HRB400-Cr-RE > HRB400-Cr > HRB400 at every chloride level, underscoring that Cr/RE incorporation suppresses interfacial charge transfer and strengthens long-term durability[29]. Importantly, HRB400-Cr-RE preserves the highest Rp even at elevated chloride concentrations, a trend that mirrors the superior pitting resistance observed in the potentiodynamic curves of Fig. 3.

Mott-Schottky curves

The semiconducting nature of the passive films can be rationalised by molecular-orbital theory. The space-charge capacitance C (F cm^{-2}) is related to the applied potential E (V) and the flat-band potential E_{fb} (V) by the Mott–Schottky (M-S) relation in Eq. (2):

$$\frac{1}{C^2} = \pm \frac{2}{\varepsilon\varepsilon_0 eN}\left(E - E_{fb} - \frac{kT}{e}\right) \quad (2)$$

Alternatively, the thickness of the space charge layer (L) can be calculated using Eq. (3):

$$L = \frac{2\varepsilon\varepsilon_0}{eN_D}\sqrt{\left(E - E_{fb} - \frac{kT}{e}\right)} \quad (3)$$

Where the capacitance of the surface oxide, C (F cm^{-2}), scales with the applied bias E (V) and the flat-band potential E_{fb} (V), the latter extracted from the x-intercept of the linear Mott–Schottky trace. Its dielectric response is expressed through the vacuum permittivity (ε_0) and the film's relative permittivity ε at 298 K, while thermal and electronic contributions are captured by the Boltzmann constant (k), absolute temperature T, and elementary charge (e = 1.6 × 10^{-19} C). Positive slopes in the Mott–Schottky plot denote n-type behaviour; negative slopes signify p-type conduction.

Figure 5 presents the Mott–Schottky curves for the three steels in SCP with increasing NaCl. All films exhibit a linear, positive-slope region between −0.5 and +0.35 V (vs SCE), characteristic of n-type semiconductors[30]. The donor densities derived from these slopes are plotted in Fig. 5e. Elevated N_D values point to non-stoichiometric, highly defective oxide lattices[31]. Irrespective of chloride concentration, HRB400-Cr-RE consistently registers the lowest N_D; for example, at 1 wt% NaCl, N_D falls from 6.19 × 10^{19} cm^{-3} for HRB400 to 2.68 × 10^{19} cm^{-3} for HRB400-Cr-RE. Based on Eq. 4, Cr/RE alloying thickens the space-charge layer and thereby suppresses ingress of aggressive species, explaining the markedly higher corrosion resistance observed in both EIS and potentiodynamic tests.

Corrosion behavior in Cl-containing simulated concrete pore fluids

Figure 6 summarises the corrosion rates after immersion for three days. HRB400 exhibited the highest initial rate (0.12 mm/a), whereas both Cr- and Cr/RE-modified steels registered markedly lower values. Over this period, the Cr/RE alloy reduced the corrosion rate by ~67% relative to conventional HRB400, reflecting the rapid establishment of a dense, Cr/RE-rich oxide barrier[32]. This finding aligns with the potentiodynamic, EIS, and Mott–Schottky data. As immersion continued, the rates for all steels declined progressively as additional corrosion products accumulated and further stabilised the surface[33].

Corrosion behavior in early stage

Early-stage corrosion initiation was examined by immersing the three rebar grades in saturated $Ca(OH)_2$ + 3.5 wt% NaCl for 10 and 30 min, followed by FE-SEM/EDS analysis. Figures 7, 8 display the FE-SEM micrographs and corresponding EDS analyses of inclusions in the three rebar specimens both

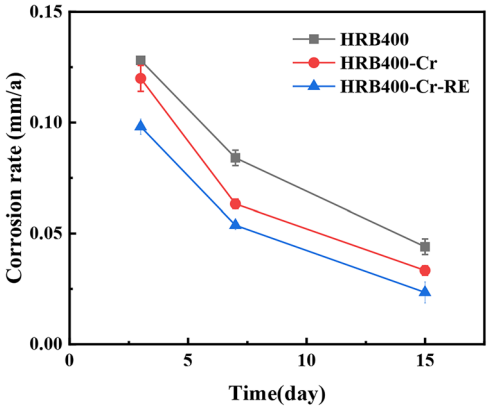


Fig. 6 | Corrosion rates in SCP solution containing 1 wt% NaCl. Comparison of corrosion rates of three samples.

prior to and following immersion in 3.5 wt.% NaCl-containing saturated $Ca(OH)_2$ solution for 10 and 30 minutes, respectively.

HRB400 (Fig. 7a_1) contains irregular Al_2O_3–MnO–CaO complex inclusions. After 10 min these inclusions begin to dissolve, opening micro-gaps at the inclusion/matrix interface and triggering localized pitting. Density-functional calculations of work functions (experimentally corroborated[34]) establish the corrosion-propensity sequence RE–Al–O–S > Fe > Al_2O_3; the less-noble Fe matrix is therefore preferentially dissolved adjacent to Al_2O_3-rich particles, progressively widening the interface gap.

In HRB400-Cr-RE (Fig. 7c_1), MnS inclusions dissolve completely within 10 min, whereas RE–Al–O–S particles dissolve only partially. The higher electrochemical potential of RE–Al–O relative to MnS drives this selective dissolution, generating micropits that nucleate preferentially at inclusion corners and edges. Matrix attack is minimal at this stage, confirming that early corrosion is governed by inclusion chemistry rather than by the steel matrix itself.

After 30 min of exposure (Fig. 8a_1), HRB400 showed extensive matrix dissolution around the Al_2O_3–MnO–CaO inclusions, producing large cavity-type pits surrounded by thick corrosion products. Two synergistic effects drive this acceleration: (i) the interfacial gap that forms as the inclusion dissolves[34] and (ii) lattice distortion arising from the thermal-expansion mismatch between the inclusion and the Fe matrix[35]. Both factors promote rapid Fe dissolution and pit nucleation.

In HRB400-Cr, the (Ca,Mg,Al)-O complex inclusions remain essentially intact after 30 min (Fig. 8b_1). Their anodic character and limited chloride adsorption preclude dissolution or detachment, so pitting continues to be governed by the earlier MnS dissolution rather than by these oxide particles. HRB400-Cr-RE contains RE–Al–O particles initially encapsulated by MnS layers (Fig. 8c_1). After 30 min the MnS shell is completely removed and the RE–Al–O cores begin to dissolve; some inclusions have vanished entirely, leaving stable pits. Comparison of Fig. 8b_2 (30 min) with earlier 10-min micrographs confirms that both pit depth and corrosion-product coverage increase markedly with immersion time.

The results demonstrate that inclusions act as preferential corrosion initiation sites, with localized matrix dissolution occurring around MnS or Al_2O_3 particles where surface potential is lower. Once pits nucleate, accumulated corrosion products form occluded cells that accelerate propagation. In contrast, the (Ca,Mg,Al)–O complex inclusions in HRB400-Cr remain intact after 30 min immersion—neither dissolving nor detaching—thereby suppressing pit initiation. This stability underpins the superior corrosion resistance of HRB400-Cr, a conclusion corroborated by both potentiodynamic and EIS data. In HRB400-Cr-RE, early attack originates from RE–Al–O–MnS and RE–Al–O inclusions. Rare-earth addition markedly lowers sulfur content, eliminating discrete MnS; The residual RE-rich inclusions—whether RE–Al–O or MnS-sheathed RE–Al–O—act as robust barriers that suppress pit propagation, substantially enhancing the global corrosion performance of HRB400-Cr-RE.

Observation of corrosion morphology after immersion

Figure 9 presents a comprehensive comparison of the corrosion morphologies exhibited by the three rebar grades after 3, 7 and 15 days of exposure to chloride-contaminated simulated concrete pore solution. During the initial three-day period, HRB400-Cr-RE maintained an almost uniform surface with only sparse, shallow pits, signifying minimal corrosive attack, whereas HRB400-Cr already displayed distinct chloride-induced pits highlighted by yellow arrows, and HRB400 exhibited pronounced rust accumulation. This morphological hierarchy remained consistent with the weight-loss trends shown in Fig. 6, corroborating the superior chloride resistance of HRB400-Cr-RE. As immersion was extended to 7 and 15 days, HRB400 underwent a progressive transition from localized pitting to generalized corrosion, accompanied by a marked increase in both the volume and particle size of corrosion products and a corresponding decline in structural stability. In contrast, HRB400-Cr exhibited moderate product build-up with enlarging yet still discrete pits, while HRB400-Cr-RE retained a thin, adherent oxide layer that intermittently exposed the underlying substrate without excessive scaling. The sustained protection afforded to HRB400-Cr-RE is attributed to the stabilizing effect of Cr/RE micro-alloying, which promotes a dense, self-healing passive film and effectively suppresses excessive corrosion-product formation throughout the entire exposure period.

Scanning-laser confocal microscopy (CLSM) was used to assess the effect of Cr/RE alloying on the early-stage pitting of HRB400 after 3 days in SCP solutions of varying chloride content; the resulting three-dimensional topographies are shown in Figs. 10 and S2~ S4 (*Supplementary material*). In 0.5 wt% NaCl pitting dominated for every steel, yet HRB400 suffered the most aggressive attack, producing pits as deep as 13.25 μm, whereas HRB400-Cr-RE displayed the best resistance, with maximum depths limited to 2.38 μm and the smallest overall damaged area. To quantify the geometry of these pits, their aspect ratio K (depth/width) was calculated; higher K denotes sharper, more vertically developed cavities. HRB400 yielded the largest K (up to 0.35, Fig. 10a1), signalling narrow, deeply penetrating pits, while the alloyed grades exhibited significantly lower values. Depth-distribution histograms (Fig. 10a2–c2) corroborate this trend: HRB400 pits centred near 2 μm, HRB400-Cr near 1 μm and HRB400-Cr-RE mostly below 0.5 μm, confirming that Cr and, especially, Cr/RE additions markedly suppress pit propagation.

With chloride concentration rising from 0.5 wt% to 3.5 wt%, all three steels experienced a pronounced mitigation of pitting attack. HRB400, for instance, saw its deepest cavities drop from 13.25 μm to 3.45 μm, while the bulk of pits clustered between 0.2 and 0.6 μm. Correspondingly, the maximum aspect ratio (K) climbed to ~1.2, signalling a shift toward lateral enlargement and an increasingly uniform corrosion front. Throughout this transition, HRB400-Cr-RE retained both the lowest peak K-values and the shallowest depth distributions, underscoring the efficacy of Cr/RE micro-alloying in suppressing pit initiation during the earliest stages of chloride exposure.

Characterization of passive film

XPS was employed to determine the composition and chemical state of the surface oxides formed on the three rebars surface after 7 days in SCP solutions with varying NaCl concentrations; the corresponding spectra (Fig. 11 and Figs. S5–7) and peak assignments are compiled in Table S1. In every specimen the Fe $2p_3/_2$ envelope could be deconvoluted into three contributions at 706.1 ± 0.5 eV (Fe^0), 709.9 ± 0.2 eV (Fe^{2+}) and 711.7 ± 0.1 eV (Fe^{3+}), consistent with a bilayer passive film in which Fe^{3+} species dominate the outer layer while Fe^{2+} species reside closer to the metal[36]. The O 1 s region exhibited two components at 529.5 ± 0.5 eV (O^{2-} in Fe_3O_4/FeO) and 531.4 ± 0.3 eV (OH^- in FeOOH), confirming the presence of both oxides and hydroxides[37–39].

In HRB400-Cr and HRB400-Cr-RE, the Cr $2p_3/_2$ envelope exhibits a faint yet sharp component at 576.9 ± 0.1 eV, attributed to Cr_2O_3 (Fig. 11c), indicating that the limited bulk Cr segregates preferentially into the passive layer. HRB400-Cr-RE additionally shows signatures at 889.4 eV (CeO_2) and 835.5 eV (La_2O_3) (Fig. 11d), confirming the integration of rare-earth oxides

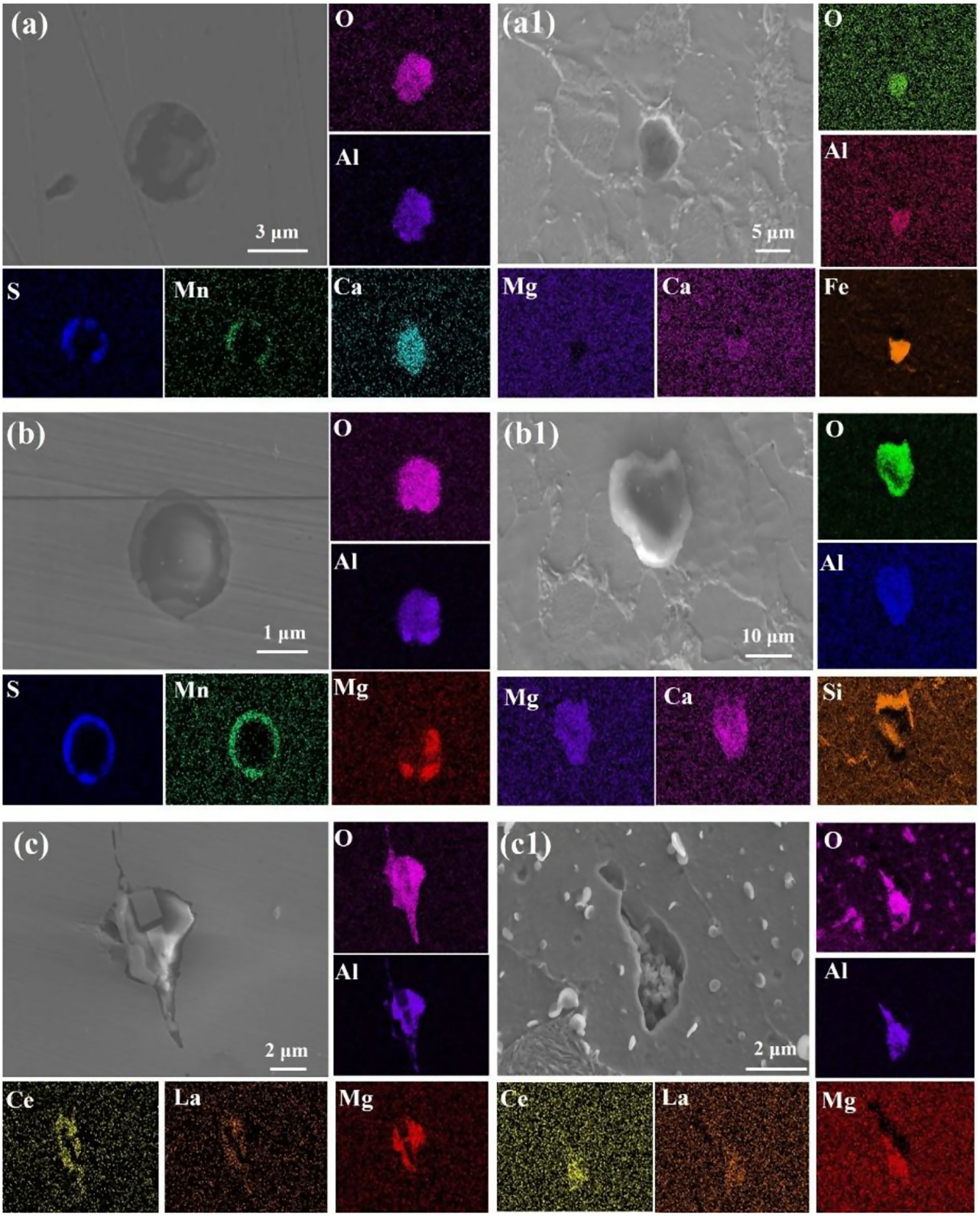


Fig. 7 | SEM and EDS of inclusions before and after 10-minute immersion in SCP solution containing 3.5 wt% NaCl. a–a1 HRB400. **b–b1** HRB400-Cr. **c–c1** HRB400-Cr-RE.

within the film. These Ce- and La-rich domains effectively block defects and hinder ion migration, accounting for the markedly higher corrosion resistance of the RE-bearing alloy.

In alkaline SCP media, Fe(II) is the dominant valence species responsible for sustaining passivity on carbon-steel rebars. The RE-modified steel shows a smaller slope based on the Mott–Schottky plots, implying a lower donor density and more stable passive film. Considering that the surface film contains mixed Fe(II)/Fe(III) oxides (Fig. 11), the response likely arises from the semiconductive and ionic transport behavior of a Fe–Ce–O layer rather than an ideal depletion-type space-charge

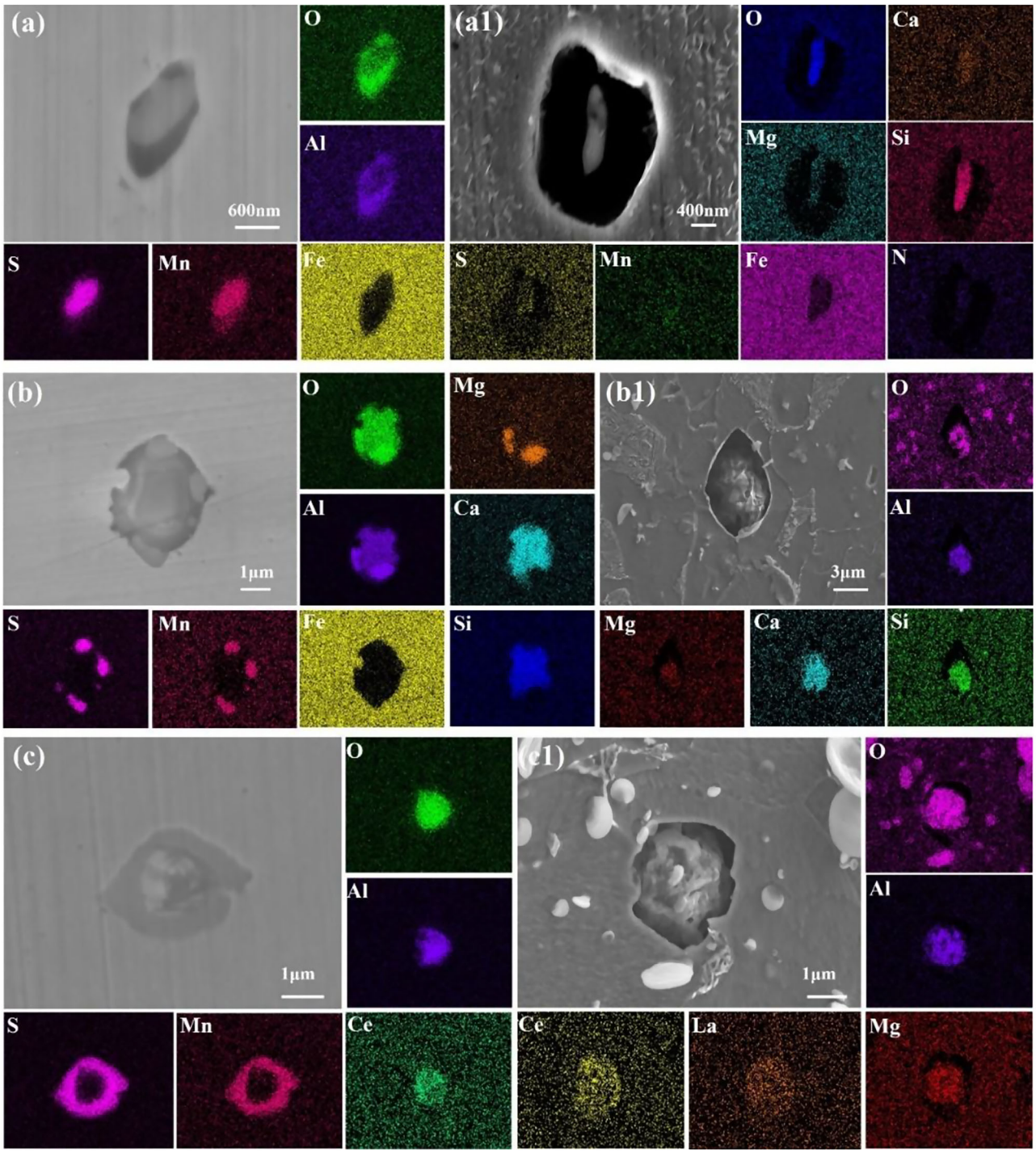


Fig. 8 | SEM and EDS of inclusions before and after 30 min immersion in SCP solution containing 3.5 wt% NaCl. a–a1 HRB400, **b–b1** HRB400-Cr, **c–c1** HRB400-Cr-RE.

capacitance, consistent with literature on RE-doped iron oxides. Consequently, the Fe(II)/Fe(III) ratio is widely accepted as a reliable indicator of passive-film integrity, with lower values signalling diminished corrosion resistance[40]. Figure S8 shows that, irrespective of chloride level, this ratio follows the sequence HRB400-Cr-RE > HRB400-Cr > HRB400, confirming that Cr/RE micro-alloying suppresses electron transfer from Fe(II) to Fe(III) and thereby strengthens the protective film. Figure 12 further reveals that the ratio decreases monotonically with increasing NaCl concentration, reflecting progressive dissolution of iron oxides/hydroxides and the consequent deterioration of passivity at higher chloride contents.

Discussions

Chloride ions are particularly destructive because they not only trigger and accelerate corrosion of the reinforcing steel but also directly attack the concrete matrix. Under cyclic wet–dry conditions, crystallisation of chloride salts generates internal stresses that propagate extensive cracking and ultimately compromise structural integrity[41]. After cement hydration, the high alkalinity (pH > 12) produced by $Ca(OH)_2$ normally drives formation of a passive film on the steel surface that prevents corrosion. Penetration of chlorides, however, destabilises this film and establishes electrochemical cells between ferrite, graphite and cementitious phases, initiating rapid

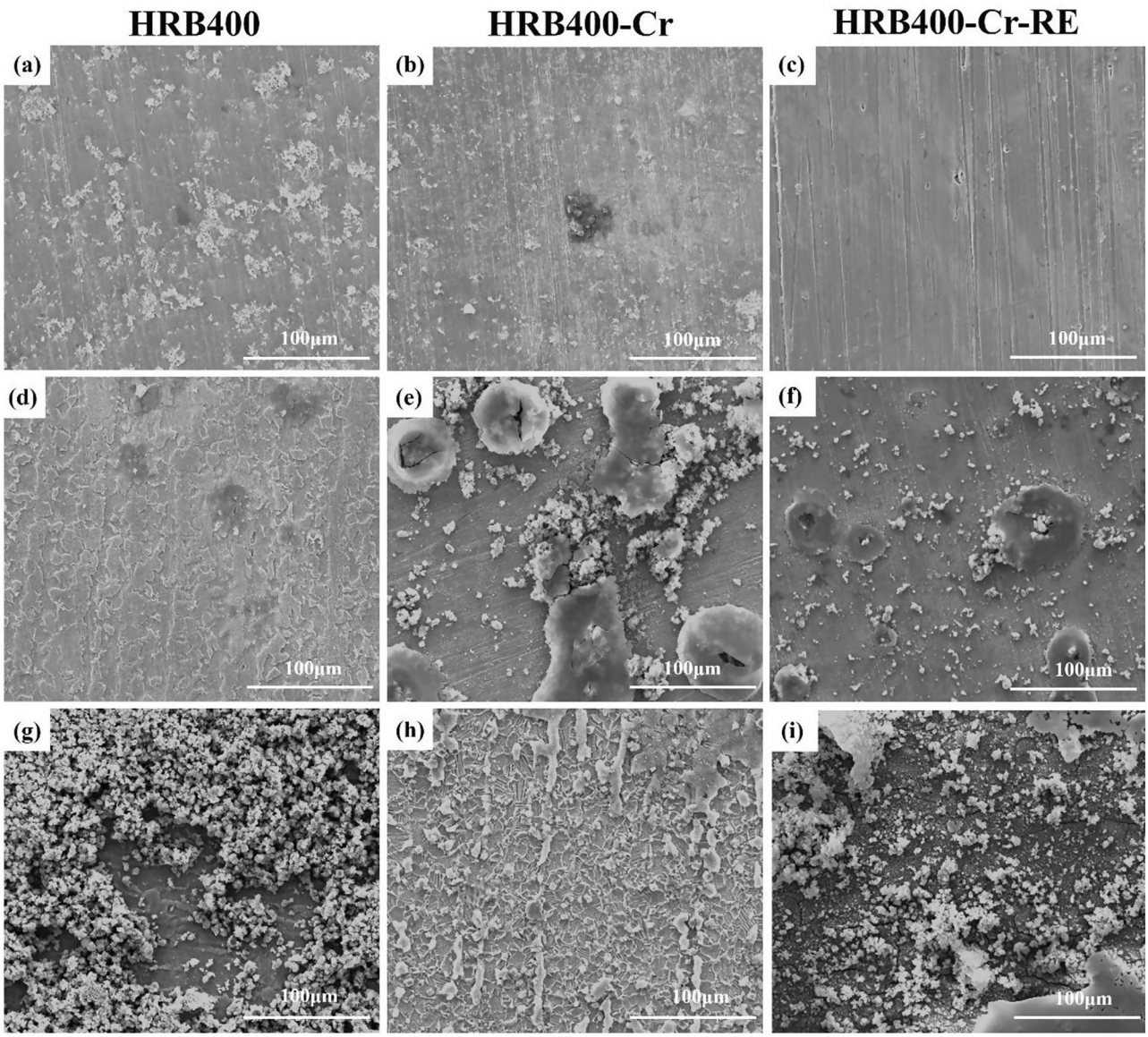


Fig. 9 | Surface morphology immersed for different times in SCP solution containing 1 wt% NaCl. a–c 3 d; **d–f** 7 d; **g–i** 15 d.

attack[42], as described by Eqs. (4, 5).

$$\text{Anodic reaction}: Fe - 2e^{-} \rightarrow Fe^{2+} \quad (4)$$

$$\text{Cathodic reaction}: O_2 + 2H_2O + 4e^{-} \rightarrow 4OH^{-} \quad (5)$$

The resulting Fe^{2+} ions are further oxidised to a succession of corrosion products based on Eqs. (6, 7)[43,44].

$$4Fe^{2+} + 8OH^{-} + O_2 + 2H_2O \rightarrow 4Fe(OH)_3 \quad (6)$$

$$2Fe(OH)_3 \rightarrow Fe_2O_3 + 3H_2O \quad (7)$$

These reactions degrade concrete in two ways[45]: (1) progressive loss of steel cross-section and (2) volumetric expansion of the corrosion products, which induces tensile stresses that crack and spall the cover. Both mechanisms weaken the steel–concrete bond, reduce composite action and threaten safety and serviceability. Moreover, the non-metallic inclusions formed during steelmaking—whose type and chemistry depend on the deoxidisers and alloying additions employed—are prone to dissolution in aggressive environments. Their selective attack creates localised micro-environments that facilitate ingress of aggressive ions and markedly accelerate pit initiation and propagation[46].

In HRB400 steel, the prevailing MnS and Al_2O_3–MnO–CaO inclusions act as persistent anodic sites whose dissolution initiates pitting in saturated $Ca(OH)_2$ pore solutions of varying chloride content. Progressive leaching of these inclusions creates microscopic voids while simultaneously attracting chloride ions, thereby sustaining an electrochemical current that acidifies the local environment as described by Eq. (8). In HRB400 steel, the prevailing MnS and Al_2O_3–MnO–CaO inclusions act as anodic sites that dissolve preferentially in chloride-containing pore solutions. The hydrolysis of sulphide ions produces an alkaline micro-environment, not an acidic one, as shown by Eq. (9). The local increase in pH promotes selective MnS dissolution and precipitation of $Fe(OH)_2$ at the inclusion–matrix interface, forming porous reaction layers that facilitate chloride adsorption and microcell formation. The released sulfide species combine with electrochemically generated thiosulfates to form highly soluble sulfide precipitates; within the resulting chloride-rich, sulfide-enriched and weakly acidic microlayer the passive film becomes thermodynamically unstable and ultimately ruptures[47,48]. Microscopic gaps between Al_2O_3–MnO–CaO inclusions and the steel matrix further concentrate aggressive ions, accelerating preferential Fe dissolution immediately adjacent to the inclusions, diminishing the

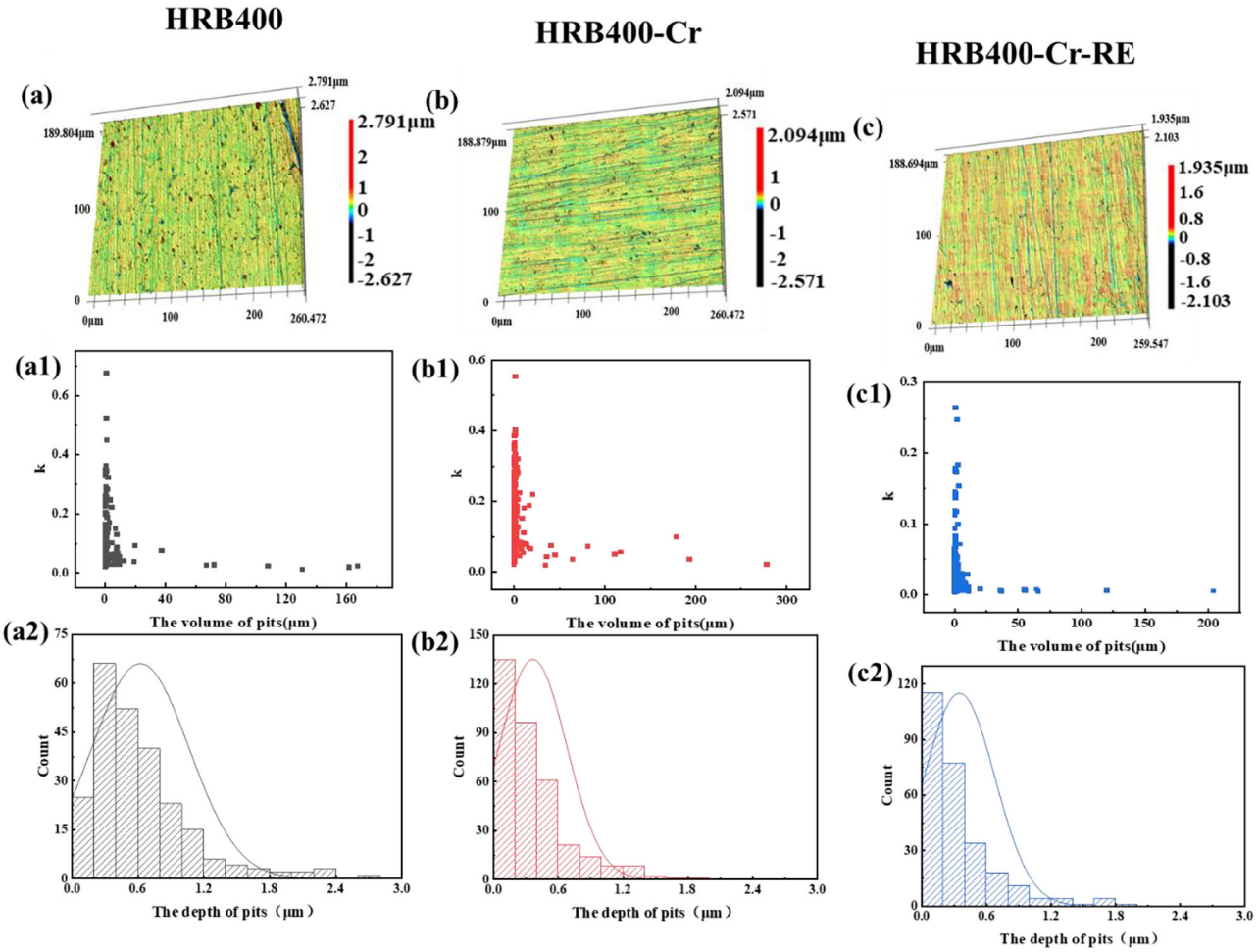


Fig. 10 | CLSM data acquired after 3 days in SCP solution with 0.5 wt% NaCl. a–c 3-D surface maps; **a_1–c_1** correlations between pit aspect ratio K and volume; **a_2–c_2** depth histograms of corrosion craters.

inclusion–matrix contact area and eventually causing complete inclusion detachment and the formation of stable pits.

$$MnS + 4H_2O \rightarrow Mn^{2+} + SO_2^- + 8H^+ + 8e^- \quad (8)$$

$$S^{2-} + H_2O \rightarrow HS^- + OH^- \quad (9)$$

HRB400-Cr contains (Ca,Mg,Al)-O complex inclusions that behave anodically. Scanning Kelvin Probe Force Microscopy (SKPFM) measurements by Zheng[49] show that these inclusions possess lower Volta potentials than the surrounding matrix[50]. Their passive film is chromium-depleted and exhibits reduced surface potential, so aggressive ions preferentially adsorb and selectively dissolve the Ca-Al-O phase, exposing fresh metal where competing pitting and repassivation events occur. Continued dissolution liberates Mg-Al-O inclusions, releasing cations (Ca^{2+}, Al^{3+}, Fe^{2+}, Ni^{2+}, Cr^{3+}) whose hydrolysis locally acidifies the solution; chloride ingress then establishes an autocatalytic acidification cycle[51] that culminates in stable pit formation once the passive film is breached[52].

In HRB400-Cr-RE, rare-earth alloying transforms Al_2O_3–MnO–CaO inclusions into RE–Al–O particles that are frequently encapsulated by MnS shells (Fig. 2c). The corrosion sequence is schematically illustrated in Fig. 13. The irregular surface of RE–Al–O–S promotes initial dissolution and local chloride accumulation, while the outer MnS layer, possessing the lowest potential, dissolves preferentially (Fig. 13b). This selective MnS removal creates interfacial gaps (Fig. 13c) that intensify attack and ultimately expose and dissolve the RE–Al–O core[53,54]. Dissolved Al^{3+} ions undergo hydrolysis and oxidation in the alkaline medium, yielding insoluble hydroxides and oxides according to the following Eq. (11).

$$(RE)AlO_3 + OH^- + 3H_2O \rightarrow RE(OH)_3(s) + Al(OH)_4^- \quad (10)$$

$$Al^{3+} + 3H_2O \rightarrow Al(OH)_3 + 3H^+ \quad (11)$$

$$Fe \rightarrow Fe^{2+} + 2e^- \quad (12)$$

$$O_2 + 2H_2O + 4e^- \rightarrow 4OH^- \quad (13)$$

Electrochemically, the inclusions act as anodic sites where electrons are released via Eqs. (10–13); these electrons are consumed by the cathodic oxygen reduction described by Eq. (13). The acidic micro-environment is supplied by H^+ ions generated during MnS dissolution, and chloride ions penetrate through surface micro-gaps to attack both inclusion and adjacent matrix[55,56]. Accumulated corrosion products impede oxygen diffusion, suppressing the cathodic reaction while chloride continues to concentrate[57,58]. Charge neutrality then forces Fe^{2+} hydrolysis, further acidifying the pit and driving rapid inclusion dissolution and pit propagation[59]. Nevertheless, the rare-earth inclusions markedly retard pit kinetics, explaining the superior corrosion resistance of HRB400-Cr-RE.

These inclusion-controlled mechanisms are fully consistent with the electrochemical data. HRB400-Cr-RE exhibits the most positive pitting potential and the lowest passive current density, the largest Nyquist arc radii, the lowest donor density, the thinnest corrosion-product layer and smallest

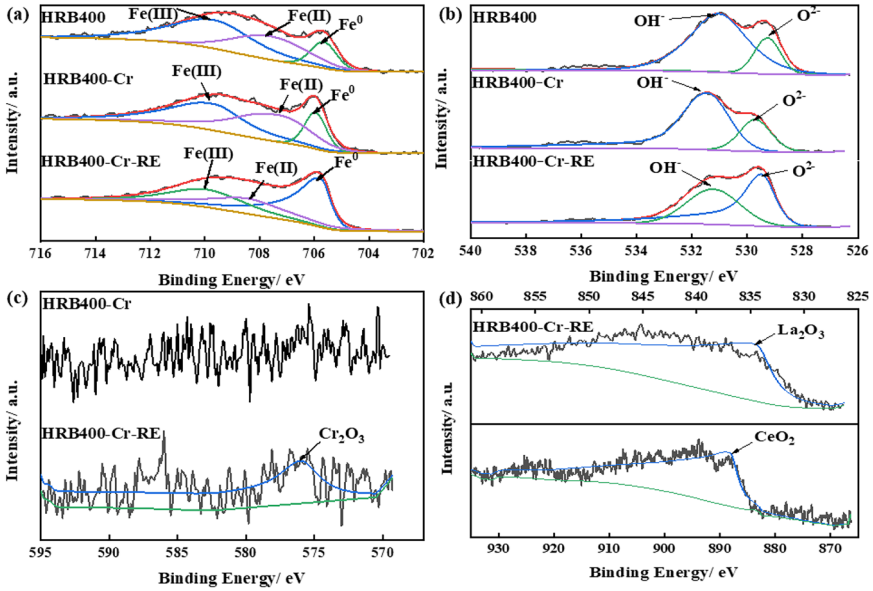


Fig. 11 | High-resolution XPS region, acquired from the oxide films of rebar samples. a Fe $2p_{3/2}$, **b** O 1 s, **c** Cr $2p_{3/2}$, and **d** Ce 3 d and La 3 d.

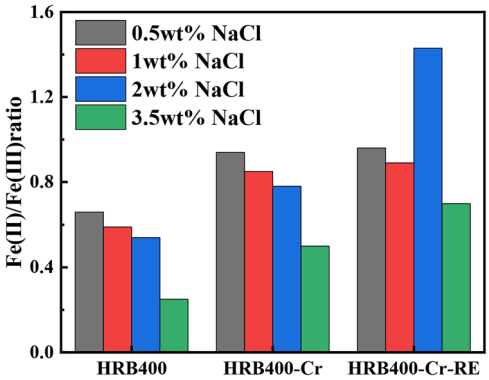


Fig. 12 | Semi-quantitative characterization of rust layer by XPS. Comparations of ratio of Fe(II) to Fe(III) of three samples.

pits after 3 d, and the shallowest pit depths and lowest aspect ratios. CLSM confirms that RE–Al–O–S inclusions—whether free-standing or MnS-encapsulated—are more effective at suppressing early corrosion than single-phase MnS, Al_2O_3–MnO–CaO or (Ca,Mg,Al)-O inclusions.

In summary, rare-earth inclusions introduced by Cr/RE micro-alloying preserve the integrity of the passive film during early exposure, maintaining a benign pH > 12 environment and yielding markedly higher passive-film resistance. Over extended periods or at high chloride concentrations, gradual inclusion dissolution and detachment erode this protection, but HRB400-Cr-RE still outperforms conventional grades because the RE-rich phases delay both pit initiation and propagation.

This study investigates the corrosion behavior of Cr/RE-modified low-alloy steel rebars in chloride-contaminated simulated concrete pore solutions. In HRB400 steel, MnS and Al_2O_3–MnO–CaO inclusions are transformed into RE–Al–O–S particles with RE addition, and corrosion initiates with the dissolution of MnS, exposing the RE–Al–O core and forming stable pits. RE-modified inclusions significantly delay pit initiation, with HRB400-Cr-RE showing the highest pitting potential, lowest passive current density, and largest Nyquist arc radius, along with the lowest corrosion rate and shallowest pits. Corrosion resistance follows the order HRB400-Cr-RE > HRB400-Cr > HRB400. As chloride concentration increases, pitting potential decreases, passive current density rises, capacitive arc radii shrink, and donor density increases, reflecting the progressive dissolution of inclusions and a corresponding decrease in overall corrosion resistance.

Methods

Materials and solution

Three low-alloy rebars samples are obtained from Xinxing Ductile Iron Pipe Co., Ltd in China. After compositional verification in Table 3 by Optical emission spectrometer (OES), the alloys underwent hot-rolling processing. The rare earth elements incorporated in this study are cerium (Ce) and lanthanum (La). For the introduction of these RE elements, a La-Ce alloy (60 wt% La) was employed. For metallographic analysis, we prepared $10 \times 10 \times 3$ mm^3 specimens from rolled plates, etched them in 4 vol.% nital solution for 5 seconds, then immediately rinsed with ethanol and dried with compressed air prior to microstructural characterization.

Analysis and characterization of inclusions

For microstructural analysis, $10 \times 10 \times 10$ mm^3 steel samples were sequentially ground (400–4000 grit SiC paper) and mechanically polished, followed by ultrasonic cleaning in anhydrous ethanol and air drying. We performed comprehensive characterization using a JEOL JSM-7100F FE-SEM/EDS system at 30 kV, 10 nA probe current, and 10 mm working distance, analyzing ≥20 inclusions per sample for statistical reliability. Quantitative inclusion assessment (size distribution, area fraction) was conducted using an ASPEX automated analyzer to ensure rigorous microstructural evaluation.

Electrochemical measurements

Electrochemical measurements were performed using a conventional three-electrode system, with the steel reinforcement specimens ($10 \times 10 \times 5$ mm^3) serving as the working electrode (1 cm^2 exposed area). A platinum sheet and saturated calomel electrode (SCE) were employed as the counter and reference electrodes, respectively. Prior to testing, the working electrodes were encapsulated in epoxy resin to ensure geometric consistency, then sequentially ground to 2000-grit using silicon carbide abrasive paper. To eliminate surface contaminants, the specimens were ultrasonically cleaned in anhydrous ethanol and deionized water, followed by drying under controlled nitrogen flow to prevent oxidation.

The electrochemical tests were conducted in a simulated concrete pore solution (saturated $Ca(OH)_2$, pH ~12.5) at 25 ± 1 °C. The chloride concentration was systematically varied (0–3.5 wt% NaCl) to evaluate corrosion behavior under different aggressive conditions. Open-circuit potential (OCP) monitoring was first performed until stabilization (±2 mV/h variation), followed by potentiodynamic polarization tests. The polarization scans were initiated at -250 mV vs. OCP (cathodic direction)

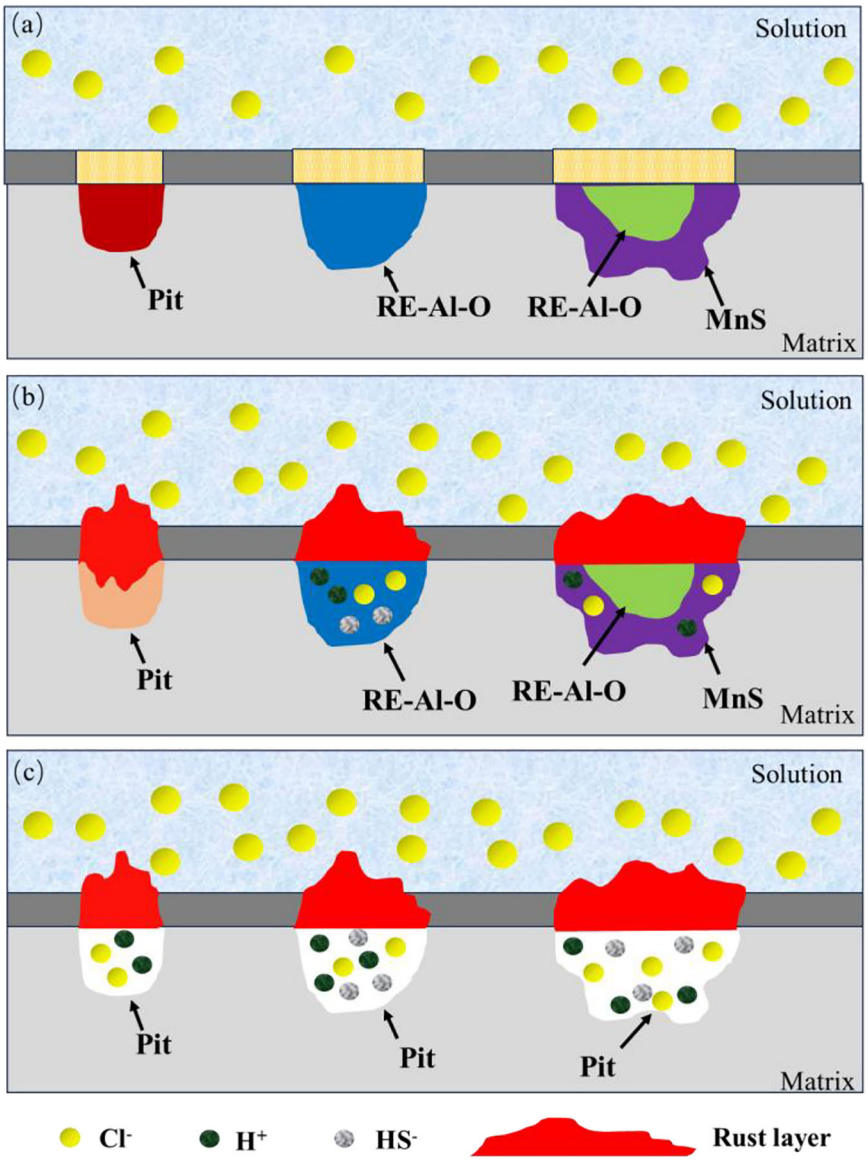


Fig. 13 | Schematic of inclusion-induced corrosion mechanism. a Corrosion initiated at inclusion-matrix interfaces. **b** Selective dissolution of MnS, exposing the inner RE–Al–O layer. **c** Expansion of the gap between the RE–Al–O inclusions and steel matrix, leading to further corrosion and dissolution of RE–Al–O.

Table 3 | Chemical compositions of three rebar samples (wt.%)

	C	Si	Mn	P	S	Cr	Ce/La	Fe
HRB400	0.24	0.5	1.5	0.006	0.06	–	–	Bal.
HRB400-Cr	0.12	0.5	1.4	0.007	0.08	0.5	–	Bal.
HRB400-Cr -RE	0.12	0.5	1.5	0.007	0.08	0.5	0.018	Bal.

and terminated at an anodic current density of 1 mA/cm², with a scan rate of 0.33 mV/s.

Electrochemical impedance spectroscopy (EIS) measurements were subsequently carried out at the stabilized OCP over a frequency range of 100 kHz to 10 mHz after immersion for 0.5 h, applying a 10 mV sinusoidal perturbation. The acquired impedance spectra were analyzed using ZsimpWin software, fitting the data to appropriate equivalent circuit models to extract key electrochemical parameters.

Immersion test

Cylindrical steel samples (Ø20-22 mm × 50 ± 2 mm) were prepared by abrasive cleaning and precisely weighed (±0.0001 g). Accelerated corrosion tests were performed in chloride-contaminated simulated concrete pore solution (3-15 days). Post-immersion, corrosion products were removed using a standardized derusting solution (1:1 HCl:H_2O with hexamethylenetetramine inhibitor) via 1-minute ultrasonic cleaning, followed by ethanol rinsing and drying. Corrosion rates were calculated from mass loss measurements using Eq. (14):

$$V_{corr} = \frac{87600 \times \Delta w}{S \times T \times \rho} \tag{14}$$

Δw denotes the mass change (g) of the coupon after immersion, S its exposed geometric area (cm²), T the exposure duration (h), and ρ the steel density (g cm^{-3}).

Corrosion morphology was examined using FE-SEM on gold-sputtered specimens (10 × 10 × 3 mm³). 3D surface topography of derusted samples was quantified by Confocal laser scanning microscope (CLSM). XPS analysis (Kratos AXIS Ultra DLD) with Al Kα radiation characterized passivation films. Survey (1 eV step) and high-resolution (0.1 eV step) scans were energy-referenced to C 1 s (285.0 eV). Quantitative composition analysis was performed via spectral deconvolution (Avantage software) using standard reference spectra.

Data availability

The datasets generated and/or analyzed during the current study are not publicly available due to the data are part of an ongoing study but may be available from the corresponding author on reasonable request.

Received: 16 August 2025; Accepted: 19 January 2026;
Published online: 07 February 2026

References

1. Fu, C., Fang, D., Ye, H., Huang, L. & Wang, J. Bond degradation of non-uniformly corroded steel rebars in concrete. *Eng. Struct.* **226**, 111392 (2021).
2. Hornbostel, K., Angst, U. M., Elsener, B., Larsen, C. K. & Geiker, M. R. Influence of mortar resistivity on the rate-limiting step of chloride-induced macro-cell corrosion of reinforcing steel. *Corros. Sci.* **110**, 46–56 (2016).
3. Zhao, Y., Dong, J., Ding, H. & Jin, W. Shape of corrosion-induced cracks in recycled aggregate concrete. *Corros. Sci.* **98**, 310–317 (2015).
4. Mundra, S., Criado, M., Bernal, S. A. & Provis, J. L. Chloride-induced corrosion of steel rebars in simulated pore solutions of alkali-activated concretes. *Cem. Concr. Res.* **100**, 385–397 (2017).
5. Berrocal, C. G., Lundgren, K. & Löfgren, I. Corrosion of steel bars embedded in fibre reinforced concrete under chloride attack: state of the art. *Cem. Concr. Res.* **80**, 69–85 (2016).
6. Tang, S. W., Yao, Y., Andrade, C. & Li, Z. Recent durability studies on concrete structure. *Cem. Concr. Res.* **78**, 143–154 (2015).
7. Tran, D. T., Lee, H.-S., Singh, J. K. & Lee, D.-E. Corrosion prevention of steel rebar embedded in the cement mortar under accelerated conditions: combined effects of phosphate and chloride ions. *Constr. Build. Mater.* **365**, 130042 (2023).
8. Xiong, C. et al. Preparation of phytic acid conversion coating and corrosion protection performances for steel in chlorinated simulated concrete pore solution. *Corros. Sci.* **139**, 275–288 (2018).
9. Liu, Q. et al. A novel green reinforcement corrosion inhibitor extracted from waste Platanus acerifolia leaves. *Constr. Build. Mater.* **260**, 119695 (2020).
10. Moffatt, E. G., Thomas, M. D. & Fahim, A. Performance of high-volume fly ash concrete in marine environment. *Cem. Concr. Res.* **102**, 127–135 (2017).
11. Aguirre-Guerrero, A. M., Robayo-Salazar, R. A. & de Gutiérrez, R. M. A novel geopolymer application: coatings to protect reinforced concrete against corrosion. *Appl. Clay Sci.* **135**, 437–446 (2017).
12. Calero, J. C., Llorca, M. C. & Terradillos, P. G. Influence of different ways of chloride contamination on the efficiency of cathodic protection applied on structural reinforced concrete elements. *J. Electroanal. Chem.* **793**, 8–17 (2017).
13. Etteyeb, N., Dhouibi, L., Takenouti, H. & Triki, E. Protection of reinforcement steel corrosion by phenylphosphonic acid pre-treatment PART II: tests in mortar medium. *Cem. Concr. Compos.* **65**, 94–100 (2016).
14. Etteyeb, N., Dhouibi, L., Takenouti, H. & Triki, E. Protection of reinforcement steel corrosion by phenyl phosphonic acid pre-treatment PART I: tests in solutions simulating the electrolyte in the pores of fresh concrete. *Cem. Concr. Compos.* **55**, 241–249 (2015).
15. Liu, M., Cheng, X., Li, X. & Lu, T. J. Corrosion behavior of low-Cr steel rebars in alkaline solutions with different pH in the presence of chlorides. *J. Electroanal. Chem.* **803**, 40–50 (2017).
16. Roventi, G., Bellezze, T., Giuliani, G. & Conti, C. Corrosion resistance of galvanized steel reinforcements in carbonated concrete: Effect of wet–dry cycles in tap water and in chloride solution on the passivating layer. *Cem. Concr. Res.* **65**, 76–84 (2014).
17. Du, F. et al. Chloride ions migration and induced reinforcement corrosion in concrete with cracks: a comparative study of current acceleration and natural marine exposure. *Constr. Build. Mater.* **263**, 120099 (2020).
18. Ida, N., Nishimoto, M., Muto, I. & Sugawara, Y. Role of MnS in the intergranular corrosion and depassivation of sensitized Type 304 stainless steel. *npj Mater. Degrad.* **8**, 2 (2024).
19. Kuah, K. X. & Blackwood, D. J. Investigating molybdenum's sulphur scavenging ability for MoS2 formation in preventing pitting corrosion of stainless steels. *npj Mater. Degrad.* **7**, 80 (2023).
20. Li, Z. et al. Role of segregation behavior of Cu and Sb in the region of inclusions on initial corrosion. *npj Mater. Degrad.* **7**, 29 (2023).
21. Liu, P., Zhang, Q.-H., Watanabe, Y., Shoji, T. & Cao, F.-H. A critical review of the recent advances in inclusion-triggered localized corrosion in steel. *npj Mater. Degrad.* **6**, 81 (2022).
22. Saito, H., Nishimoto, M. & Muto, I. Pitting corrosion characteristics of sintered Type 316 L stainless steel: relationship between pores and MnS. *npj Mater. Degrad.* **8**, 61 (2024).
23. Reformatskaya, I., Rodionova, I., Beilin, Y. A., Nisel'Son, L. & Podobaev, A. The effect of nonmetal inclusions and microstructure on local corrosion of carbon and low-alloyed steels. *Prot. Met.* **40**, 447–452 (2004).
24. Szklarska-Smialowska, Z. Influence of sulfide inclusions on the pitting corrosion of steels. *Corrosion* **28**, 388–396 (1972).
25. Domizzi, G., Anteri, G. & Ovejero-Garcıa, J. Influence of sulphur content and inclusion distribution on the hydrogen induced blister cracking in pressure vessel and pipeline steels. *Corros. Sci.* **43**, 325–339 (2001).
26. Liu, M., Cheng, X., Li, X., Jin, Z. & Liu, H. Corrosion behavior of Cr modified HRB400 steel rebar in simulated concrete pore solution. *Constr. Build. Mater.* **93**, 884–890 (2015).
27. Liu, M., Cheng, X. & Li, X. Corrosion propagation behavior of low-Cr steel Rebars in simulated concrete environments. *Int. J. Electrochem. Sci.* **14**, 726–742 (2019).
28. Yuan, X., Wang, X., Cao, Y. & Yang, H. Natural passivation behavior and its influence on chloride-induced corrosion resistance of stainless steel in simulated concrete pore solution. *J. Mater. Res. Technol.* **9**, 12378–12390 (2020).
29. Yan, L., Song, G.-L., Wang, Z. & Zheng, D. Crevice corrosion of steel rebar in chloride-contaminated concrete. *Constr. Build. Mater.* **296**, 123587 (2021).
30. Liu, M., Cheng, X., Li, X., Pan, Y. & Li, J. Effect of Cr on the passive film formation mechanism of steel rebar in saturated calcium hydroxide solution. *Appl. Surf. Sci.* **389**, 1182–1191 (2016).
31. Xu, L. et al. Structural characteristics and chloride intrusion mechanism of passive film. *Corros. Sci.* **207**, 110563 (2022).
32. Liu, M. et al. Indoor accelerated corrosion test and marine field test of corrosion-resistant low-alloy steel rebars. *Constr. Build. Mater.* **5**, 87–99 (2016).
33. Yang, D., Yan, C., Zhang, J., Liu, S. & Li, J. Chloride threshold value and initial corrosion time of steel bars in concrete exposed to saline soil environments. *Constr. Build. Mater.* **267**, 120979 (2021).
34. Zhang, X. et al. Effects of niobium and rare earth elements on microstructure and initial marine corrosion behavior of low-alloy steels. *Appl. Surf. Sci.* **475**, 83–93 (2019).
35. Wei, J., Dong, J., Ke, W. & He, X. Influence of inclusions on early corrosion development of ultra-low carbon bainitic steel in NaCl solution. *Corrosion* **71**, 1467–1480 (2015).
36. An, H. et al. High corrosion resistance film on rebar by cerium modification. *J. Mater. Sci. Technol.* **64**, 73–84 (2021).
37. Chen, T. et al. Insights into the role of the Cr and rare element in improving the corrosion resistance of HRB400 rebars in simulated SO2-polluted marine environment. *J. Build. Eng.* **97**, 110807 (2024).
38. Chen, T. et al. Deterioration mechanism of passivation behavior of ductile iron induced by shrinkage defects in simulated concrete pore solution. *J. Mater. Sci. Technol.* **236**, 136–149 (2025).

39. Chen, T. et al. The coupling mechanism of shrinkage defects and graphite on the corrosion resistance of ductile iron. *Corros. Sci.* **227**, 111798 (2024).
40. Dong, B. et al. Corrosion failure analysis of low alloy steel and carbon steel rebar in tropical marine atmospheric environment: outdoor exposure and indoor test. *Eng. Fail. Anal.* **129**, 105720 (2021).
41. Al-Amoudi, O. S. B., Rasheeduzzafar, A., Maslehuddin, M. & Abduljauwad, S. Influence of sulfate ions on chloride-induced reinforcement corrosion in portland and blended cement concretes. *Cem. Concr. Aggreg.* **16**, 3–11 (1994).
42. Zhang, Z., Zhang, P., Niu, G., Ding, C. & Wu, H. Enhanced passivation of Cr-modified rebar in simulated seawater sea-sand concrete chloride environment. *J. Mater. Res. Technol.* **35**, 7051–7064 (2025).
43. Chen, T. et al. Understanding the mechanism of shrinkage defects on corrosion kinetics of ductile iron. *Corros. Sci.* **256**, 113168 (2025).
44. Chen, T. et al. Comparative study on corrosion resistance of carbon steel and ductile iron: Implications for the development of corrosion-resistant steels. *Corros. Sci.* **255**, 113127 (2025).
45. Chen, T. et al. Corrosion behavior of 650 MPa high strength low alloy steel in industrial polluted environment containing different concentration of Cl-. *Int. J. Miner. Metall. Mater.* https://doi.org/10.1007/s12613-025-3135-5 (2025).
46. Williams, D. E., Kilburn, M. R., Cliff, J. & Waterhouse, G. I. Composition changes around sulphide inclusions in stainless steels, and implications for the initiation of pitting corrosion. *Corros. Sci.* **52**, 3702–3716 (2010).
47. Montemor, M., Simoes, A. & Ferreira, M. Chloride-induced corrosion on reinforcing steel: from the fundamentals to the monitoring techniques. *Cem. Concr. Compos.* **25**, 491–502 (2003).
48. Dehwah, H. A., Austin, S. A. & Maslehuddin, M. Chloride-induced reinforcement corrosion in blended cement concretes exposed to chloride-sulphate environments. *Mag. Concr. Res.* **54**, 355–364 (2002).
49. Zheng, S., Li, C., Qi, Y., Chen, L. & Chen, C. Mechanism of (Mg, Al, Ca)-oxide inclusion-induced pitting corrosion in 316L stainless steel exposed to sulphur environments containing chloride ion. *Corros. Sci.* **67**, 20–31 (2013).
50. Chen, T. et al. Exploring trace Mg microalloying impact on corrosion mechanism of X70 Steel. *Metall. Mater. Trans. A* **56**, 3518–3535 (2025).
51. Batista, W., Louvisse, A., Mattos, O. & Sathler, L. The electrochemical behaviour of INCOLOY 800 and AISI 304 steel in solutions that are similar to those within occluded corrosion cells. *Corros. Sci.* **28**, 759–768 (1988).
52. Mikhailov, A. S., Scully, J. R. & Hudson, J. L. Nonequilibrium collective phenomena in the onset of pitting corrosion. *Surf. Sci.* **603**, 1912–1921 (2009).
53. Chen, T. et al. Insights into multiple coupling mechanisms of $SO4^{2-}$/Cl^- and Cr/RE elements on the corrosion resistance of rebar in simulated carbonated concrete pore solution. *Constr. Build. Mater.* **485**, 141957 (2025).
54. Chen, T. et al. Assessing the durability of low-alloy rebars in China plateau environment by outdoor exposure and on-site online monitoring. *Constr. Build. Mater.* **468**, 140475 (2025).
55. Liu, C. et al. Effect of inclusions modified by rare earth elements (Ce, La) on localized marine corrosion in Q460NH weathering steel. *Corros. Sci.* **129**, 82–90 (2017).
56. Liu, C. et al. Role of Al_2O_3 inclusions on the localized corrosion of Q460NH weathering steel in marine environment. *Corros. Sci.* **138**, 96–104 (2018).
57. Che, Z. C. et al. Role of Te-RE alloying on the passive film and pitting corrosion behavior of 316L stainless steel. *Corros. Sci.* **240**, 112457 (2024).
58. Liu, C. et al. New insights into the mechanism of localised corrosion induced by TiN-containing inclusions in high strength low alloy steel. *J. Mater. Sci. Technol.* **124**, 141–149 (2022).
59. Liu, C. et al. Influence of rare earth metals on mechanisms of localised corrosion induced by inclusions in Zr-Ti deoxidised low alloy steel. *Corros. Sci.* **166**, 108463 (2020).

Acknowledgements

The authors acknowledge the National Natural Science Foundation of China (No. 52374323).

Author contributions

R.Z.: writing-original draft & editing; T.C.: data curation; L.H.: investigation; W.Y., C.G., X.Z., and X.C.: methodology; C.L.: methodology, supervision, writing-review & editing; X.L.: supervision, methodology. All authors read and approved the final version.

Competing interests

X.L. serves as editor of this journal and had no role in the peer review or decision to publish this manuscript. The remaining authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41529-026-00746-3.

Correspondence and requests for materials should be addressed to Chao Liu or Xiaogang Li.

Reprints and permissions information is available at http://www.nature.com/reprints

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

© The Author(s) 2026